

Pre-registration by an isolated agent: what it protects against, and what it does not

Reza Nehzati, Ph.D.

VMC MAR COM Inc. DBA HeyDonto, Knoxville, TN, United States

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Abstract

Language-model agents can now write and execute clinical data analyses. The obvious hazard is not incompetence but flexibility: a system that sees the data before fixing its plan can produce a defensible-looking finding from almost any dataset. We test a protocol designed to remove that flexibility, on three sealed oncology cohorts. A generator agent receives only schema and documentation — form names, column names, row counts, a data dictionary, and no cell value from any outcome field — and must emit a complete pre-registered plan: hypotheses, positive controls with pre-declared failure consequences, integrity gates, an exhaustive exclusion ledger, a power statement, and mechanical verdict rules. The plan is SHA-256 hashed before any outcome is read, and the exposure table is frozen and hashed before outcomes are merged, so post-outcome tuning is mechanically detectable rather than merely discouraged. Across three sealed cohorts the system produced three distinct behaviours. On the first (247 stage-IV NSCLC patients treated with anti-PD-(L)1) it returned POWER-LIMITED-NUL rather than a finding: the pre-specified multimodal comparison was structurally underpowered at $n = 73$, and the plan’s own rule required saying so. On the second (ACRIN 6668, 166 analysed patients, 125 events), a structurally isolated generator produced a plan that recovered the trial’s designed endpoint — post-treatment FDG peak SUV predicting shorter survival, HR 1.202 per doubling, permutation $p = 0.0005$ — having identified four analysis-destroying traps from schema alone, including immortal-time bias and a vital-status code that silently undercounts deaths. On the third (46 anti-PD-1 patients with imaging but no released outcome) it declined outright, returning **feasible: false** with the reasoning that outcome missingness was 100% and the exposure was a *constant* — every patient received the drug that defines the collection, so no contrast exists — and it identified a covariate leak in the metadata we had supplied without noticing it ourselves. The protocol’s limits are as instructive as its successes. In the first cohort a pre-registered hard gate failed and was reinterpreted after the fact. In the second, an implementation error let a control’s hazard ratio be reported as the primary result while all eight gates passed. We conclude that pre-registration constrains *analytic* discretion effectively and *implementation* correctness not at all, and that the two require different defences. Two further gaps are now visible: neither defence prevents a small-sample estimate being reported as though it were stable, and neither prevents a plan from measuring a confounded quantity with complete procedural correctness. The last is the most serious, because the protocol’s own machinery reports success throughout.

1 Introduction

An agent that can read a data dictionary, write analysis code and execute it can also, without any intent to deceive, arrive at a result by trying things until something works. The number of defensible-looking choices in a survival analysis — time origin, exclusion rules, transformation, cutpoint, covariate set, handling of ambiguous status codes — is large enough that a sufficiently persistent search will find a significant one.

Pre-registration is the standard answer, and it transfers to this setting with one modification that matters: the plan can be written by a system that has provably not seen the outcomes, and the ordering can be enforced by a hash rather than by a declaration.

We ask two questions. Does such a system produce a plan a competent analyst would accept? And does it refuse to manufacture findings the data cannot support? We report three cohorts — one confirmation, one power-limited null, one outright refusal — and we report the protocol’s failures at equal length.

2 Protocol

1. **Recon package.** The generator receives a single JSON file: every form’s name, column names and row counts; the trial’s data dictionary; public facts about the cohort; and join counts. It contains no cell value from any outcome, measurement or clinical field.
2. **Isolation.** In the second and third cohorts the generator was a separate agent whose readable surface was a directory containing only that file. The location of the actual data was never disclosed to it.
3. **Plan.** The generator emits hypotheses with named fields, positive controls each with an explicit pre-declared failure consequence, numbered integrity gates with expected values, an exhaustive exclusion ledger, a power statement computed before execution, multiplicity handling, mechanical verdict rules, and a list of conclusions it will refuse to draw.
4. **Hash before labels.** The coordinator SHA-256 hashes the plan and records the timestamp before reading any outcome value.
5. **Staged execution with an exposure freeze.** Structural gates run first on keys and counts only. The exposure table is then built, written and hashed. Only then may outcome forms be opened — enforced in the second cohort by a loader that raises if asked for an outcome form before the freeze. A later gate re-hashes the exposure table and compares, making post-outcome modification of the predictor mechanically detectable.
6. **Mechanical verdict.** Verdict rules are evaluated from the numbers, in a declared precedence order, with no discretion at the final step.

3 Cohort 1 — refusal under low power

247 stage-IV NSCLC patients treated with anti-PD-(L)1, with RECIST response, survival, PD-L1 immunohistochemistry slides, targeted panel sequencing and CT radiomics. The original authors’ analysis code was placed under a self-imposed firewall for the whole arm.

3.1 What the system did

Eleven gates were declared and ten passed, including hash-before-label ordering, exclusion accounting that reconciles per endpoint, absence of NaN or constant prediction vectors with 100% out-of-fold coverage, and permutation nulls centred at 0.494–0.495 across every model family. That null centring is what makes the accompanying p -values worth reading.

The pre-declared positive control — PD-L1 tumour proportion score, *on its own* — predicted response at AUROC **0.747** (permutation $p = 1.0 \times 10^{-4}$, $n = 246$), clearing its declared bar of AUROC > 0.55 at $p < 0.05$. The multivariable clinical baseline that the primary comparison is measured against — PD-L1 plus tumour mutational burden plus clinical variables — reached AUROC **0.745** on the same 246 patients (0.742 as the mean over cross-validation repeats).

Corrected 2026-08-17. This paragraph originally attributed the 0.747 to the multivariable baseline. It is the single-variable positive control; the baseline model is 0.745. The number was taken from the evidence but assigned to the wrong model, and the error survived a verification pass that compared values without checking which field produced them. It does not affect the primary comparison below, which is a paired contrast against the multivariable baseline on the complete-case subset. Found by the scoped re-verification described in `papers/verify_claims.py`.

The pre-specified primary comparison asked whether adding imaging and pathology to that baseline improved it. On the 73 patients with every modality present, the change was -0.036 (95% CI -0.220 to $+0.150$, one-sided $p = 0.638$). The plan’s rule mapped this to **POWER-LIMITED-NULL**: a bootstrap interval half-width of 0.185 cannot distinguish a useful gain from a harmful one, so no finding was claimed in either direction.

Three further modality additions, each pre-declared, were all negative and none individually significant: radiomics -0.016 , pathology-global -0.019 , pathology-focal -0.056 . Four negative estimates out of four is more informative than any single interval, and the honest reading is that at these sample sizes added parameters cost more than they return.

Holm correction across the model family left two survivors of six. The system reported the four that did not survive as not claimable, including one at raw $p = 0.018$.

3.2 An unexpected observation

Embeddings from a general-purpose pathology foundation model predicted a PD-L1 tumour proportion score of $\geq 50\%$ at AUROC **0.870** (95% CI 0.811–0.922, $n = 163$, 67 positive) with no immunohistochemistry-specific training. This was a competence control, not a hypothesis; we report it because it was surprising, and note that a second definition of the same quantity gives 0.856, so the figure is definition-dependent.

3.3 The protocol’s first failure

A pre-registered *hard* gate failed. The control asked whether segmentation-derived tumour volume predicted survival, with the stated basis “proves I can read the imaging”. Observed: C-index **0.5035**, Wald $p = 0.698$, on 187 patients with 111 events. Flat.

The asymmetry in the plan is the problem. Its pathology control carried an explicit remedy written in advance — “failure $\Rightarrow$ report pipeline not competent, not evidence of absence”. The imaging control carried none: it was a bare pass/fail gate. So when it failed there was no pre-registered instruction, and a radiomics model built on that same imaging was one of only two Holm survivors.

A defence is available and reasonable — the control tested whether target-lesion segmentations approximate *total* tumour burden, a different and questionable premise, and three other controls passed. But it is a **post-hoc reinterpretation of a pre-registered control**, and calling it anything else would defeat the purpose of the exercise. The instructive part is that pre-registration surfaced a control whose basis was mis-specified, which is what pre-registration is for.

4 Cohort 2 — confirmation under isolation

ACRIN 6668 / RTOG 0235: FDG-PET before and after definitive chemoradiotherapy in stage III NSCLC, released with clinical data under CC BY 3.0. Here the generator was structurally isolated as described in §2, and produced a 493-line plan hashed at `752b7462fce56899...cb74e6770` before any outcome was read.

4.1 What it caught from schema alone

- **Immortal time.** The exposure is measured 8–20 weeks after treatment, so measuring survival from registration would guarantee survival to exposure. It set the time origin at the post-treatment scan and made $\min(T) \geq 0$ a gate.
- **A vital-status code meaning “dead, date of death unknown”.** The natural `event = (status == 2)` silently undercounts deaths. It treated the code as an event and made the opposite coding a *condition* of confirming.
- **Cutoff forking.** One form carries 1,185 rows keyed by an SUV inclusion threshold — roughly five analyses per patient. It locked to a single threshold and forbade comparison across them.
- **An undocumented uniqueness assumption.** Its primary exposure required one row per patient in a form where the recon package never said so. It demanded verification and pre-specified a duplicate-resolution rule. The rule fired: 244 rows over 243 patients.

It also pre-declared fifteen conclusions it would refuse to draw, and forbade cutpoint searches, subgroup fishing, and endpoint substitution.

4.2 Result

Primary: OS on $\log_2$ post-treatment peak SUV	HR per doubling	1.202
	95% CI	1.100–1.315
	permutation p ($B = 2000$)	0.0005
	analysed / events	166 / 125
PC0 exposure validity (outcome-free)	post below pre	PASS (94.4%)
PC1 progression $\rightarrow$ death	HR 2.463	PASS ($p = 3.8 \times 10^{-5}$)
PC2 performance status	HR 1.382	PASS-WEAK ($p = 0.072$)
PC3 weight loss $\geq 5\%$	26 exposed	UNINFORMATIVE-BY-N

Ten gates passed, the exclusion ledger reconciled exactly ($251 = 166 + 85$), the permutation null was centred and calibrated (mean z 0.034, SD 0.982, rejection rate 0.042), and the exposure hash was unchanged from the Stage-1 freeze. All seven pre-declared robustness tests agreed in direction and survived Holm correction, including the adjusted model (1.214), a fixed threshold with no search permitted (1.789), and the opposite death-coding (1.202).

Verdict: CONFIRMED — the trial’s designed endpoint, recovered by a plan written in ignorance of the data’s contents.

Two controls did not fully corroborate. Performance status was directionally right but not significant; weight loss was declared uninformative because 26 patients met the exposure threshold against a pre-declared minimum of 30. Both had been *anticipated*: trial eligibility screens performance status, so range restriction was given its own outcome branch in advance rather than explained afterwards.

The plan also forced an uncomfortable number into the open. **106 of 180 exposure values were at or below the floor** — complete metabolic response — so 59% of the cohort sits at one tied value and the association is carried by the minority with residual uptake. A plan written after seeing the data would have had every opportunity not to mention this.

4.3 The protocol’s second failure, and the more important one

An earlier execution reported the primary as HR 1.382 with 95% CI 1.050–1.124: a point estimate outside its own confidence interval, which is impossible. Inside the control blocks the coordinator had reused two variable names, clobbering the primary’s values before the verdict logic read them. The verdict was therefore briefly evaluated using a *control’s* hazard ratio.

All eight of the plan’s gates passed while this was true. The gates checked recombination, joins, cutoff inflation, leakage, exclusion accounting, permutation calibration, exposure coverage and outcome completeness. None of them could see that the number being reported was not the number that had been computed. What caught it was an internally impossible pair of values, noticed by a human reading the output.

The conclusion survives correction. That it was momentarily correct by accident is the point.

5 Cohort 3 — refusal when no analysis exists

The two cohorts above both had endpoints. A protocol that only ever produces analyses has not demonstrated the second half of the claim, so we gave the isolated generator a cohort with none: 46 non-small-cell lung cancer patients treated with anti-PD-1, 677 imaging series across CT, PET and three “other” series, pre-treatment and one follow-up timepoint. The archive released images and a retrieval manifest and nothing else — no response assessment, no survival, no demographics, no stage, no lesion annotation.

It was asked whether a valid pre-registered analysis was possible and to write one if so. It returned `feasible: false` in 567 lines.

5.1 Its reasoning

Two defects, each independently fatal. Outcome missingness is 46 of 46: not a missing-data problem, because there is no observed stratum to model from and no estimand can be defined. And **the exposure is a constant** — the collection is *defined* by every patient having received anti-PD-1, so there is no contrast, no comparator, and no released regimen or line variation to recover one. The second point is the sharper of the two and we had not articulated it ourselves.

It nonetheless computed the power ceiling in advance, so that a later “repaired” version could not quietly overreach: at $n = 46$ with roughly 32 events the minimum detectable hazard ratio is about 2.7, which puts signature *discovery* permanently out of reach and admits only a confirmatory test of a pre-frozen external signature.

5.2 The shortcuts it named and refused

Most valuable is what it identified as tempting rather than merely absent. The gravest is a pseudo-RECIST endpoint built from the single follow-up scan: a volumetric change looks like a reader-free quantitative outcome and would plausibly survive review. It is disqualified *specifically in this cohort* because under anti-PD-1 early enlargement can reflect immune infiltration rather than growth, so the surrogate misclassifies in a direction correlated with the biology under study — bias that points at the hypothesis rather than adding noise.

It also flagged the three “other” series as bait for optical character recognition, since dose reports and console captures are where rendered text survives, and noted this would both cover too few patients and attempt to defeat de-identification; free-text protocol headers as a source of grep-able outcome labels such as “restaging” or “progression”, which are scheduling artefacts with no denominator; and a maximum of 36 series per patient as reconstruction redundancy rather than 36 timepoints.

And it found a leak we had introduced. The recon package listed the most common protocol names, among them 1.2 BMI 0-25 FDG Body and BMI_25to40_PETCTBody_CBM. These leak a coarse body-habitus covariate through weight-stratified PET protocols, with an obesity-paradox literature hook ready to receive it. We had included those strings without noticing. It declined to use them — protocol-selection labels are not measurements, the bands do not partition, they are observed only in PET-imaged patients, and they reconstruct a variable the custodian chose not to release.

5.3 What it offered instead

Rather than pad the refusal, it proposed an outcome-free study explicitly fenced as a different question — about the instrument rather than the patient: a header-only harmonisability audit with a pre-declared rule that if the largest homogeneous acquisition stratum falls below 15 patients, “this collection cannot support radiomics” *is* the published result. Its extension turns the reconstruction redundancy into a repeatability estimate using automatic non-lesion reference organs, which removes the reader-discretion problem outright rather than mitigating it.

It closed by conditioning its own verdict: the refusal rests on the recon package, and does not survive a custodian demonstrating that a patient-level clinical supplement exists.

6 Four occasions when the discipline actually cost something

The three cohorts above show a protocol behaving well. They do not show it *binding*, because in none of them did a pre-declared rule prevent a conclusion we wanted. Four later runs did, and they are worth stating because a constraint that never bites is decoration. Two concern the protocol’s mechanics; two concern its limits, and those are the more useful.

6.1 A threshold that would have been convenient to move

A five-encoder survey of site-leakage in whole-slide features reused verdict thresholds fixed and hashed in an earlier run, before that run’s own results were read. One encoder returned a mean methylation inflation of +0.0173 where the other three histology encoders gave +0.0606 to +0.0849.

Had the deciding rule been written that day it would naturally have been a magnitude bar — and at any bar near 0.05 that encoder fails, turning a clean four-for-four result into a mixed one. The rule that actually existed keyed on *how many of six targets* inflated rather than by how much, so it fired, and the fivefold spread went into the limitations where it belongs. We could not have chosen either framing honestly after seeing the number. The pre-registration made the choice for us, and it happened to choose the stricter-in-kind option.

6.2 A pattern we are not allowed to claim

The same survey produced its most interesting observation by accident: relative inflation falls as site-disjoint capability rises, Spearman $\rho = -0.80$ across five encoders. It looks like a law when plotted.

It is not reported as a finding, because it was noticed *after* all five arms were read, and at $n = 5$ no rank correlation can reach $p < 0.05$ by that test — the observed p is 0.104. Testing a hypothesis on the data that suggested it is precisely the flexibility this paper is about. It is recorded in the run’s own output under a field whose value is the string POST-HOC OBSERVATION, NOT A TESTED FINDING, and what it licenses is a pre-registration on encoders not yet used, not a conclusion.

This is the uncomfortable case. The most publishable observation in that run is the one we are least entitled to, and the protocol offers no way to have it.

6.3 And a gate that caught a statistic rather than a bug

A later run re-measured an external cohort separation whose value had been reported as 0.8768. A reproduction gate pinned that number as a hard expectation. Re-run with a different random seed the same quantity came out 0.9240, and the gate halted the run.

Nothing was broken. The seed sets the cross-validation split, and on 76 samples with roughly fifteen per fold a few reassignments move balanced accuracy several points; measured over 25 splits the value is 0.9176 ± 0.0247 . The originally reported 0.8768 was the pre-declared seed — so not chosen to flatter anything — but it sat 1.6 standard deviations below its own mean and had been quoted as a fixed quantity. The earlier report’s conclusion survived; its headline contrast did not.

The gate was written to detect a coding difference and detected a statistical one. This is a third failure mode, distinct from both of the above: not analytic flexibility, not implementation error, but *treating a noisy estimate as a point*. It suggests a fourth defence — report distributions over resampling wherever n is small — which none of our pre-registrations required.

6.4 A plan that measured the wrong thing, competently

The protocol was applied to a question about drug resistance: in 187 lung cancer cell lines with 450k methylation and matched dose–response, are lines resistant to standard therapy differentially sensitive to chromatin-targeting drugs? The plan was hashed before any half-maximal inhibitory concentration was joined to any methylation value.

It returned a clean, significant, pre-registered answer: resistant lines are *cross*-resistant, Spearman $\rho = +0.651$, permutation $p = 0.0005$, $n = 187$. Under the declared rule that is THE-SIS_CONTRADICTED.

Then a control we added afterwards destroyed it. Both indices were mean z -scored $\ln\text{IC}_{50}$, and cell lines differ enormously in *general* drug sensitivity — some are resistant to nearly everything. Partialling out an index built from 227 drugs in neither set takes $+0.651$ to -0.160 . The sign reverses.

We are not entitled to swap one number for the other: the post-hoc figure is not a pre-registered test and cannot replace one. What the pair licenses is the conclusion that *the design was confounded and answers nothing*.

This is the sharpest limitation we have found. A correctly executed pre-registration, whose every gate passed, produced a publishable-looking and wrong conclusion. Had we not been suspicious of how clean $+0.651$ looked, the report would have stated a manuscript’s central premise contradicted at $p = 0.0005$ in 187 cell lines. **Pre-registration constrains analytic flexibility; it does not confer construct validity.** A hashed plan that measures the wrong quantity measures the wrong quantity with excellent discipline, and nothing in the protocol detects that.

6.5 A two-sided rule that cost us the headline

The same protocol was then applied to a real intervention rather than a correlation. SMARCA4 is the ATPase of the SWI/SNF chromatin-remodelling complex, so knocking it down in an osimertinib-resistant lung cancer line is epigenetic modulation of a resistant state. The plan

asked whether that reverses the resistance transcriptional programme, and was hashed after reading only the column headers of the expression matrix.

The knockdown worked — SMARCA4 $\log_2\text{FC}$ -2.37 against a pre-declared bar of -0.5 , checked as a gate. Genes up in resistance then fell below the permutation null exactly as the thesis predicts ($p = 0.0005$). Under a one-sided rule that is a positive result.

The rule declared in advance was two-sided: reversal required resistance-up genes to fall *and* resistance-down genes to rise. The down set fell as well, so the mechanical verdict was PARTIAL. The two-sided form is the correct one — a treatment that lowers everything associated with resistance is suppressing a correlated set, not reprogramming a state, and only the up-down pattern separates those. Writing it cost nothing in advance and cost the headline when it fired.

A replication arm carried in the same plan, with no threshold declared, then inverted every sign in a second model. One model showing partial reversal and another showing the opposite is not a finding. Without that arm the report would have claimed partial reversal at $p = 0.0005$.

7 Discussion

7.1 What pre-registration bought

Concretely, in these three cohorts: a null was reported as a null with its power stated rather than as an absence of effect; four of six models were declared not claimable after multiplicity correction, including one at raw $p = 0.018$; a control was declared uninformative rather than counted as a pass; an inconvenient distributional fact about 59% of the exposure was reported; four analysis-destroying traps were identified before any data was touched; and a fixed cutpoint was used where a search would have been available and rewarding; and an entire cohort was declined rather than analysed with a surrogate endpoint that would likely have passed review.

None of that requires trusting the analyst. The hash ordering and the exposure freeze make the central claims checkable by a third party from the artefacts alone.

7.2 What it did not buy

It does not make a plan correct. The first cohort’s imaging control was mis-specified: its stated basis did not match what it actually tested. Pre-registration fixed that mistake in place where it could be seen, which is better than the alternative, but it did not prevent it.

It does not protect against implementation error. This is the finding we would most want carried forward. A pre-registration constrains which analysis you are allowed to run. It says nothing about whether the code you wrote runs that analysis. In the second cohort every gate passed while the reported primary was a different quantity from the computed one, and the discrepancy was visible only because two numbers were mutually inconsistent.

The remedy is not more pre-registration. It is a distinct class of check: internal-coherence assertions on the outputs themselves — a point estimate inside its interval, a tally summing to its total, a hash matching its earlier value. We added the first of these after the fact. A mature protocol should declare them alongside the analytic gates, because **analytic discretion and implementation correctness are different failure modes and neither defence covers the other.**

7.3 Limitations

Three cohorts and four later runs. One confirmation, one power-limited null, one outright refusal, plus four occasions where a pre-declared rule bound against our interest — including one

where a correctly executed plan produced a confidently wrong answer. That is a demonstration, not an evaluation, and it cannot support a claim about how often such a system behaves well.

Only the second generator was structurally isolated. In the first, one agent restrained itself, and a self-imposed firewall is a weaker guarantee than an inaccessible one.

The second cohort's analysis was tabular. It used core-laboratory PET assessments, not the underlying image volumes, so the claim concerns an unseen cohort and data shape rather than an unseen imaging pipeline.

Neither result is an independent replication. The second cohort's finding is the trial's own published conclusion, obtained from the same data. The claim is about process, not biology.

One coverage gate passed at exactly its threshold (180 against ≥ 180). One patient fewer would have triggered a declared fallback and changed the exposure source, which illustrates how much a pre-registered threshold can hinge on an arbitrary boundary.

The generator's isolation rests partly on its own report of which files it read, alongside the structural constraint that the data's location was never disclosed and its readable directory contained one file.

Data and code availability

Both cohorts are public: the first via Synapse, the second via The Cancer Imaging Archive under CC BY 3.0. The hashed plan, its hash and timestamp, the recon package given to the generator, the executor, the frozen exposure table with its hash, gate records, exclusion ledgers, all permutation statistics, and out-of-fold predictions are retained and available on request.

Competing interests

The author is affiliated with a commercial entity developing computational pathology methods.